

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 1 Summer 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 1

Subject Code & Name: 24AF1000ES108 - BASIC CIVIL AND MECHANICAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. Which of the following is an example of a shallow foundation?
 - a) Pile foundation
 - b) Caisson foundation
 - c) Spread Footing
 - d) Pier Foundation
2. What is the primary function of mortar in masonry?
 - a) To provide structural support
 - b) To bind brick of stones together
 - c) To improve drainage
 - d) To enhance aesthetic appeal
3. The entire assembly of styles, panels and rails is known as
 - a) Putty
 - b) Horn
 - c) Sash
 - d) Shutter
4. What is the primary function of an RCC lintel?
 - a) To provide structural support to walls
 - b) To distribute the loads over openings
 - c) To enhance aesthetic appeal
 - d) To improve insulation
5. An offset is adistance of an object measured from the survey line.
 - a) Lateral
 - b) Horizontal
 - c) Normal
 - d) Inclined

- 6Instrument is used to sight to an object in plane table surveying.
- Compass
 - U-Frame
 - Plumbing fork
 - Alidade
- 7 Horizontal distance between two points in two consecutive contours is known as.....
- Horizontal equivalent
 - Vertical equivalent
 - Contour interval
 - All of the above
- 8 Which of the following kind of energy output is obtained from a steam power plant?
- Electricity
 - Thermal energy
 - Sound energy
 - Heat energy
- 9 Which of the following is a type of thermodynamic system?
- Open System
 - Closed system
 - Isolated system
 - all of the above
- 10 Petrol engine arethan diesel engine .
- Lighter
 - Unpredictable
 - Heavier
 - None of the above
- 11 Which of the following operation is not carried out lathe machine?
- Welding
 - Facing
 - Turning
 - Parting
- 12 Sand mould & permanent mould are the part ofmanufacturing process.
- Machining
 - casting
 - welding
 - None of the above
- Q2. Solve the following.
- Define foundation along with its function. 6
 - Describe concrete along with its properties. 6
- Q3. Solve the following.
- Describe in detail Panelled doors with sketch. 6
 - Describe in detail RCC Lintel with weather shed along with sketch. 6
- Q4. Solve Any Two of the following.
- Describe in detail Metric Chain with sketch. 6
 - Describe the advantages and disadvantages of Plane Table Surveying. 6

- C) Describe 'Contouring' along with its purpose. 6
- Q5. Solve Any Two of the following.
- A) What is the function of Power Plant? Explain in brief working of thermal power plant with suitable sketch. 6
- B) Describe any six operations performed on lathe machine with suitable diagrams. 6
- C) Explain the working of four stroke petrol engine with suitable diagram. 6
- Q6. Solve Any Two of the following.
- A) Explain the law of conservation of energy. How it is related with first law of thermodynamics? 6
- B) Classify engineering material. Describe in detail properties and applications of any two non-ferrous metals. 6
- C) What is the difference between Machine and Mechanism? Explain any two types of Mechanism with suitable diagram. 6

*** End ***